DUB: Discrete Unit Back-translation for Speech Translation

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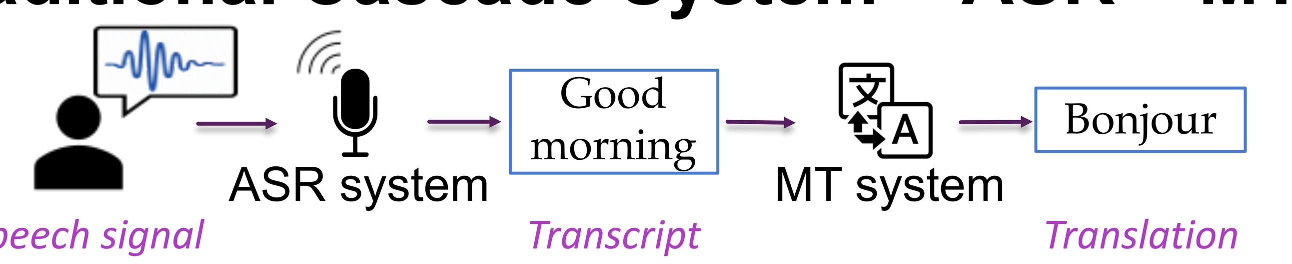
PAPER CODE

Background

➢ Task: Speech-to-text Translation (ST)

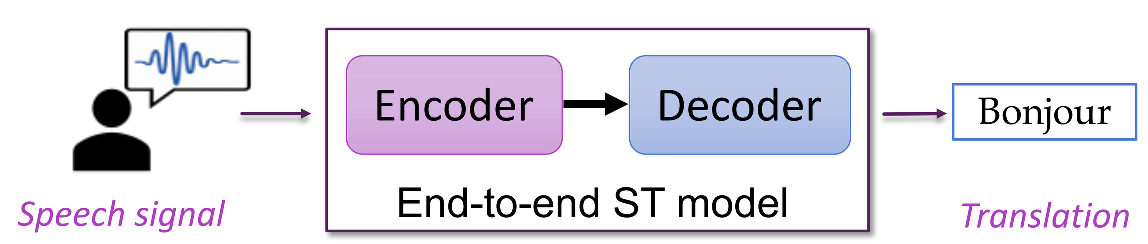
To read the ***audio signals of speech*** in one language, and translate to the ***text*** in another language. Speech translation has wide applications.

➢ Traditional Cascade System = ASR + MT



Challenges: **1.** Error propagation issue.
2. Translation for unwritten languages.

➢ End-to-end model = encoder-decoder model

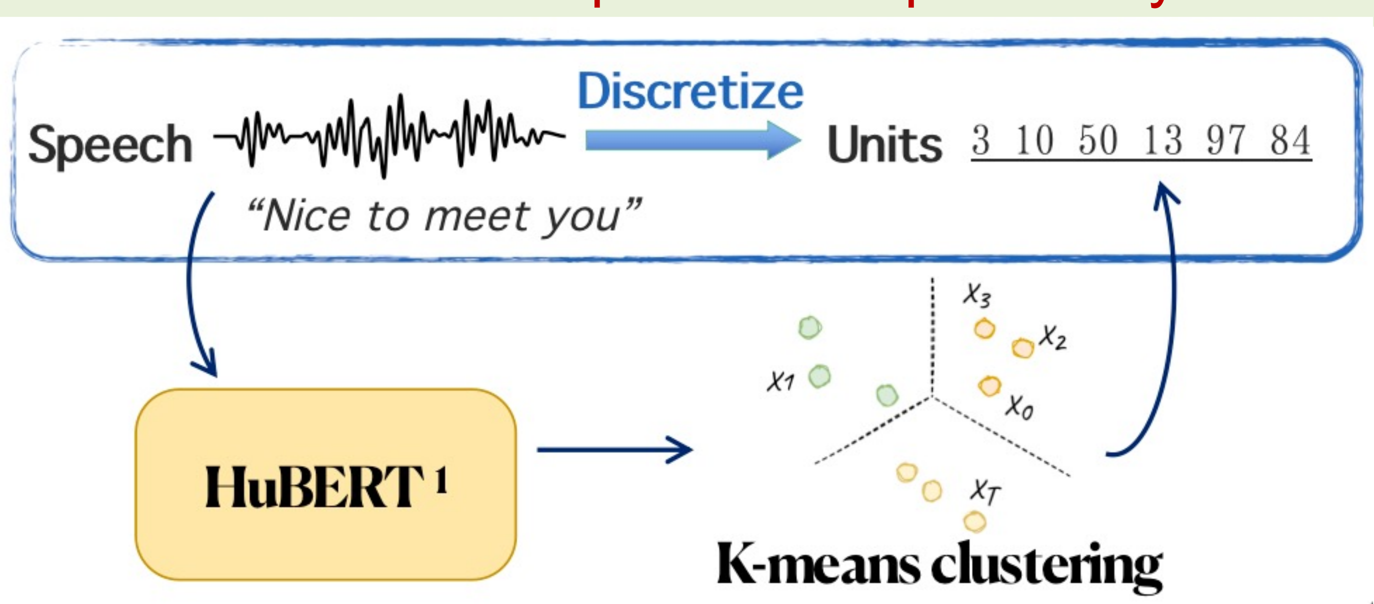


Motivations

Drawback1: **Data Scarcity** → Similar to low-resource MT
Motivation1: apply effective low-resource MT techinque to ST

↓ back-translation, pre-training

Drawback2: **Modality gap** ← symbolic gap between continuous speech and discrete text
Motivation2: discretize speech unsupervisedly



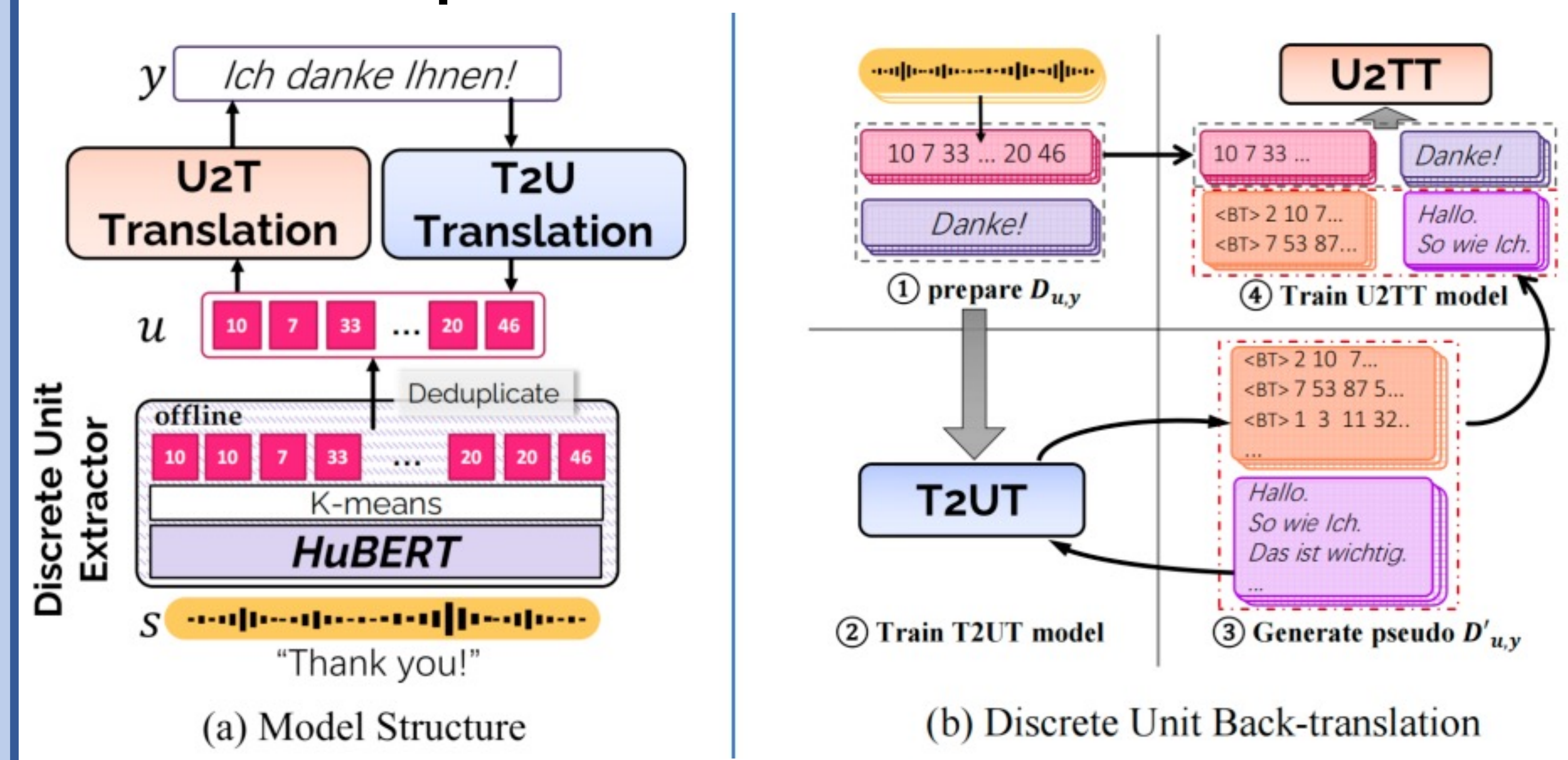
Two Questions:

Question1: **Is it better to represent speech with discrete units than with continuous features in direct ST?**

Question2: **How much benefit can useful MT techniques bring to ST?**

Framework/Process of DUB

★**CORE: Apply useful back-translation technique from MT to ST by discretizing the speech signals into unit sequences.**



(a) Model Structure (b) Discrete Unit Back-translation

➢ Framework

- ➢ discrete unit extractor: HuBERT
- ➢ Unit-to-text Translation (U2TT): Transformer
- ➢ Text-to-unit Translation (T2UT): Transformer

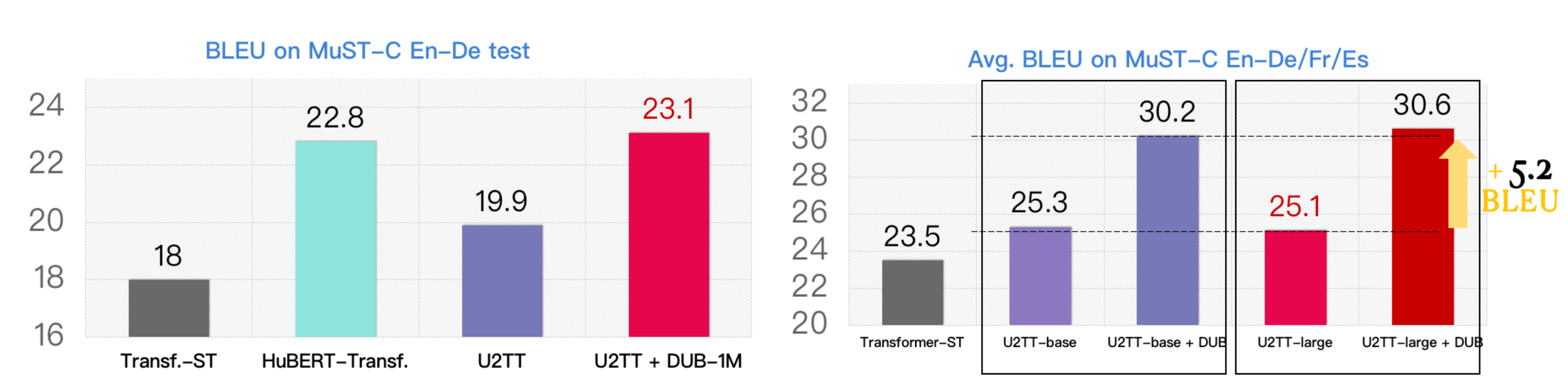
➢ Process

- ➢ same process as Back-translation in MT
- ➢ introduce text corpus and generate pseudo unit-text parallel data by T2UT to enhance U2TT model.

Main Results

➢ We conduct experiments on ST datasets

- **Discrete units are better choice for ST input*(left)***
- **DUB brings 5.2 BLEU improvement *(right)***
- **DUB is benefical for low-resource or unwritten languages *(bottom:* *results on CoVoST X-En)***

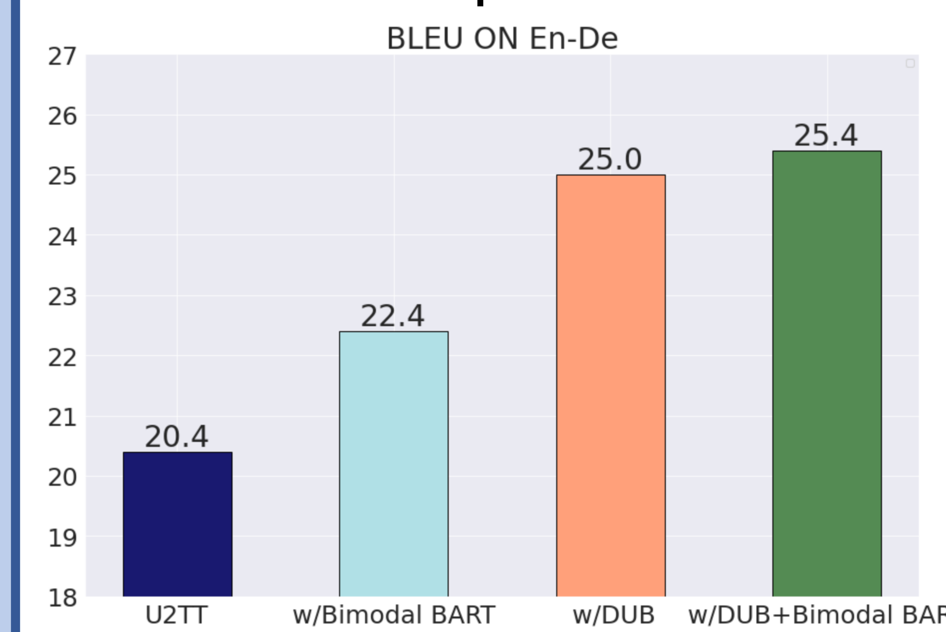


Methods	Aux. Data ASR	Aux. Data Text	Ar 2h	Sv 2h	Lv 2h	Sl 2h	Ta 2h	Ja 1h	Id 1h	Cy 2h	Avg. -
Transformer-ST†	-	-	0.3	0.2	0.1	0.3	0.3	0.3	0.4	0.3	0.3
Transformer-ST A2E†	-	-	0.6	0.6	0.4	1.2	0.1	0.2	0.3	2.6	0.8
Transformer-ST + ASR pre-train†	✓	-	4.3	2.7	2.5	3.0	0.3	1.5	2.5	2.7	2.4
Larger models based on large-scale multilingual speech, text or joint pre-training, involving more data											
XLS-R (0.4B)*	✓	-	8.1	5.3	3.1	5.3	0.0	2.0	3.3	3.4	3.8
Wav2seq (0.4B)*	✓	-	**10.5**	8.8	4.8	5.9	0.0	1.9	5.0	**5.7**	5.3
XLS-R + mBART-50 (0.7B)◇	✓	✓	3.0	**10.3**	6.0	6.6	0.2	0.6	1.4	2.5	3.8
LNA-E,D (0.7B)♠	✓	✓	3.7	5.9	4.6	4.6	**0.7**	1.7	2.9	2.8	3.4
U2TT (Large)	-	-	7.0	8.0	6.3	6.8	0.3	1.6	6.6	2.7	4.9
w/ DUB	-	✓1M	7.1	8.9	**6.9**	**7.9**	0.5	**2.1**	**7.0**	**5.7**	**5.8**

Analysis

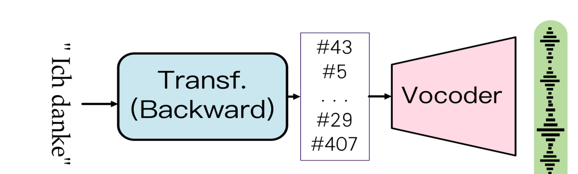
➢ We also explore effective pre-training methods like mBART

- **Bimodal BART :** Denoise large-scale corrupted discrete units and monolingual text like mBART
- Provide a pre-trained model for U2TT



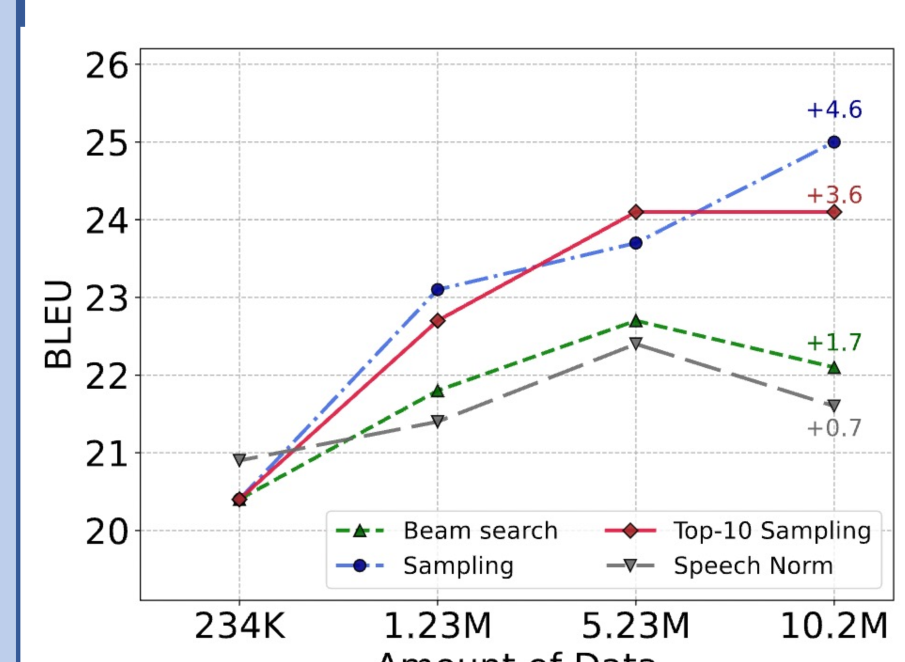
- Bimodal BART brings 2 BLEU improvement.
- Bimodal BART and DUB are **complementary.**

➢ Faithful speech can be recovered via BT units



- **Listening test on 30 BT-units-recovered speech:**
  - **22 sentences have completely identical semantics**
  - **7 sentence miss or repeat 1-2 details**
  - **1 sentence have low quality**

➢ Similar with MT: The worse units, the better performance



- **The richness and irregularity of synthesized units can better improve performance**

- *More experiments results and details, please refer our paper.*
- *Any question, email:* dongzhang22@m.fudan.edu.cn